

Chapter- Five

Introduction to Advanced Server Side Issues

FORM HANDLING

- ☞ A form is a section of an HTML document where you put user-input controls, like text boxes, check boxes, radio buttons, and pull-down lists. You use forms when you want to collect and process user input.
- ☞ ASP.NET provides important feature event handling to Web Forms. It let us to implement event-based model for our application. As a simple example, we can add a button to an ASP.NET Web Forms page and then write an event handler for the button's click event. ASP.NET Web Forms allows events on both client and server sides.
- ☞ In ASP.NET Web Forms pages, however, events associated with server controls originate on the client but are handled on the Web server by the ASP.NET.
- ☞ ASP.NET Web Forms follow a standard .NET Framework pattern for event-handler methods. All events pass two arguments: an object representing the object that raised the event, and an event object containing any event-specific information.

ASP.Net Web Form Features

- ☞ ASP.NET has multiple features and provides many tools to create and develop web applications. Here are some of the features of web forms:
 - i. Server Controls
 - ii. Master Pages
 - iii. Working with data
 - iv. Membership
 - v. Client Script and Client Frameworks
 - vi. Routing
 - vii. State Management
 - viii. Security
 - ix. Performance
 - x. Error Handling
- i. Server Control
 - ☞ It provides a vast set of server controls. These controls are like objects, and they run when they are requested and rendered to the browser. Some web

pages are similar to HTML elements like text-box, button, checkbox, and hyperlink.

ii. Master Pages

- ☞ Master Pages is responsible for the consistent layout of our web applications. It gives a proper appearance and standard to different pages.

iii. Routing

- ☞ URL routing can be configured to a web application. A request URL is a URL that a user enters in a browser to browse in a specific place.

iv. Security

- ☞ Security always plays a crucial role in software development. ASP.NET provides different configuration options and extensibility points to make our systems more secure.

ASP .NET FORM LABEL

- ☞ This control is used to display textual information on the web forms.
- ☞ It is mainly used to create caption for the other controls like: textbox.
- ☞ This is server side control, asp provides own tag to create label.

Example:

<asp: labelID = "label1" runat= "server" Text = "label1"></asp: label>

Property	Description
Access Key	It is used to set a keyboard shortcut for the label.
Tab Index	The tab order of the control.
Back Color	It is used to set the background color of the label.
Border Color	It is used to set border color of the label.
Border width	It is used to set the width of the border of the label.
Font	It is used to set the font for the label text.
Fore Color	It is used to set the color of the label text.
Text	It is used to set text to be shown for the label.
ToolTip	It displays the text when the mouse is over the label.
Visible	To set the visibility of control on the form
Height	It is used to set the height of the control.
Width	It is used to set the width of the control.

ASP .Net WEB FORM TEXTBOX

- ☞ This is an input control which is used to take user input.
- ☞ To create a textbox either we can write a code or use the drag and drop facility of visual studio IDE.
- ☞ This is server side control, run asp provides own tag to create it.

```
<asp:TextBoxID= "TextBox1" runat= "server"></asp:TextBox>
```

Property	Description
Access Key	It is used to set keyboard shortcut for the control.
Tab Index	The tab order of the control.
Back Color	It is used to set the background color of the control.
Border Color	It is used to set the border color of the control.
Border Width	It is used to set width of border of the control.
Font	It is used to set font for the control text.
Fore Color	It is used to set color of the control.
Text	It is used to set text to be shown for the control.
ToolTip	It displays the text when mouse is over the control.
Visible	To set the visibility of control on the form.
Height	It is used to set height of the control.
Width	It is used to set width of the control
Max length	It is used to set maximum number of characters that can be entered.
Read Only	It is used to make control read only.

ASP .NET DROP DOWN LIST

- ☞ The DropDownList is a web server control which is used to create an HTML Select component.
- ☞ It allows us to select an option from the dropdown list.
- ☞ It can contain any number of items ASP.Net provides a tag to create DropDownList for web application.

Example:

```
<asp:DropDownlist id= "DropDownlist1" runat = "server" DataSource = "<% databindingexpression %>"
```

```
DataTextField = "DataSourceField"
```

```
DataValueField = "DataSourceField"
```

```
AutoPostBack = "True | False"
```

```
OnSelectedIndexChanged = "OnSelectedIndexChangedMethod">
```

```
<asp: ListItem Value = "value" selected = "True | False">
```

```
Text
```

```
</asp: ListItem>
```

```
</asp: DropDownList>
```

CODE:

```
protected void Button1_click(object sender, EventArgs e)
{
    if(DropDownList1.SelectedValue == "")
    {
        Label1.Text = "Please Select a City";
    }
    else
    {
        Label1.Text = " Your Choice is:" + DropDownList1.SelectedValue;
    }
}
```

ASP .NET WEB FORMS RADIO BUTTON

- ☞ It is an input control which is used to take input from the user. It allows user to select a choice from the group of choices.

```
<asp: RadioButton ID = "RadioButton1" runat = "server" Text = "Male"
GroupName = "Gender"/>
```

Property	Description
Access key	It is used to set a keyboard shortcut for the control.
TabIndex	It is used to set Tab order of the control.
BackColor	It is used to set the background color of the control.
BorderColor	It is used to set the border color of the control.
BorderWidth	It is used to set the width of the border of the control.
Font	It is used to set the font for the control text.
ForeColor	It is used to set the color of the control text.
Text	It is used to set text to be shown for the control.
ToolTip	It display the text when the mouse is over the control.
Visible	To set visibility of control on the form.
Height	It is used to set the height of the control.
Width	It is used to set the width of the control.

GroupName	It is used to set the name of the radio button group.
-----------	---

FORM CHECKBOX

- ☞ Form CheckBox is used to get multiple inputs from the user.
- ☞ It allows user to select choices from the set of choices.
- ☞ It takes user input in yes or no format.
- ☞ It is useful when we want multiple choices from the user.

Property	Description
Access Key	It is used to set a keyboard shortcut for the control.
Tab Index	The tab order of the control
Back Color	It is used to set the background color of the control.
Border Color	It is used to set the border color of the control.
Border Width	It is used to set the border width of the control
Font	It is used to set the font for the control text.
Fore Color	It is used to set the color of the control text.
Text	It is used to set text to be shown for the control.
ToolTip	It displays the text when the mouse is over the control
Visible	To set visibility of control on the form
Height	It is used to set height of the control
Width	It is used to set width of the control.
Checked	It is used to set check state of the control either true or false.

Example:

```
<asp: checkbox ID = "CheckBox1" runat = "server" Text = "MVC" />
```

```
<asp: checkbox ID = "CheckBox2" runat = "server" Text = "CORE" />
```

```
<asp: checkbox ID = "CheckBox3" runat = "server" Text = "ASP.NET" />
```

ASP .NET FORM BOTTON

- ☞ This control is used to perform events. It is also used to submit client request to the server.

Property	Descriptions
Access Key	It is used to set a keyboard shortcut for the control.
Tab Index	It is used to set Tab order of the control.
Back Color	It is used to set the background color of the control.

Border Color	It is used to set the background color of the control.
Border Width	It is used to set the width of the border of the control.
Font	It is used to set the font for the control text.
Fore Color	It is used to set the color of the control text.
Text	It is used to set text to be shown for the control.
ToolTip	It displays the text when the mouse is over the control.
Visible	To set visibility of control on the form.
Height	It is used to set the height of the control.
Width	It is used to set the width of the control.

Example:

```
<asp: ButtonID = "Button1" runat= "server" Text = "Submit" BorderStyle = "Solid" ToolTip = "Submit"/>
```

```
<asp: Button ID = "Button1" runat = "server" Text = "Click here" OnClick = "Button1_Click"/>
```

DATABASE CONNECTIVITY

- ☞ A database connection is a facility in computer science that allows client software to communicate with database server software, whether on the same machine or not. A connection is required to send commands and receive answers.

ADO.NET

- ☞ ADO.NET is a module of .Net Framework which is used to establish a connection between application and data sources. Data sources can be such as SQL Server and XML. ADO.NET consist of classes that can be used to connect, retrieve, insert and delete data.

ADO.NET CLASSES

- ☞ Some important ADO.NET objects that are responsible for the CRUD operations is as follows:
 - ❖ **DataSet:**
Think about DataSet as a copy of a database stored in server's memory. It is used only for querying multiple SQL tables at once.
 - ❖ **SqlDataReader:**
It is used for querying data from a single SQL table.

❖ **DataTable**

Data Table is a sub item of a DataSet and represents a database table stored in the memory.

❖ **SqlConnection**

Object responsible with storing the data.

❖ **SqlCommand**

Object responsible with sending the SQL query to the server and returning the results.

❖ **SqlDataAdapter**

SqlDataAdapter is responsible with filling a DataSet with the data returned from the database.

❖ **DataReader**

This retrieve data in forward only and read only form.

❖ **DataAdapter**

This acts as a bridge between dataset and data source to load the dataset and reconcile changes made in dataset back to the source.

Web.config file with the new connection string

Syntax:

```
<connectionStrings>
```

```
    <add name = "your connectionStringName" connectionString = "Data Source =
    DatabaseServerName; Integrated Security = true; Internal Catalog =
    YouDatabaseName; uid= YourUserName; Password=yourpassword;" providerName=
    "System.Data.SqlClient"/>
```

```
</connectionStrings>
```

Example:

```
<connectionStrings>
```

```
    <add name = "myconnection" connectionString = " Data Source = miniproject;
    Integrated Security = true; Internal Catalog = MyWebDatabase" providerName =
    "System.Data.SqlClient"/>
```

```
</connectionStrings>
```

Database Connection into .cs file:

```
SqlConnection con = new
SqlConnection(ConfigurationManager.ConnectionStrings[“connectionstrings”].ToString());
```

Example:

```
SqlConnection con = new
SqlConnection(ConfigruationManager.ConnectionStrings[“myconnection”].ToString())
;
```

Here **myconnection** is connection string that we create into Web.config file.

Creating SQL Statements

1. How to create database

```
CREATE DATABASE database_neme;
USE Database_name;
```

```
Eg. CREATE DATABASE imsdbs;
USE imsdbs;
```

2. Dropping the Database (Deleting the database)

```
Syntax: DROP DATABASE database_name;
Eg. DROP DATABASE imsdbs;
```

3. How to create Table

```
CREATE TABLE Table_name
(
    ID INT NOT NULL IDENTITY(1,1),
    FirstName VARCHAR(50),
    SecondName VARCHAR(50)
)
```

4. Adding Column on Table

```
Syntax: ALTER TABLE table_name
    ADD column_name datatype;
Eg. ALTER TABLE tbl_user
    ADD Email varchar(255);
```

5. Deleting Column from Table

```
Syntax: ALTER TABLE table_name
    DROP COLUMN column_name;
```


Eg. ALTER TABLE tbl_user
DROP COLUMN Email;

6. Modifying Column (changing the data type of a column in a table)

Syntax: ALTER TABLE table_name
MODIFY COLUMN column_name datatype;

Eg. ALTER TABLE tbl_user
MODIFY COLUMN class int;

7. Deleting Table

Syntax: DROP TABLE table_name;
Eg. DROP TABLE tbl_user;

SQL Statement to insert, select, update and delete data

8. Inserting data into table

Syntax: INSERT INTO table_name(column1, column2,
column3....)VALUES(value1, value2, value3,.....);

Eg. INSERT INTO tbl_user(uid, uname, upassword)VALUES(01,'Saroj', 'singh');

9. SELECT QUERY

SELECT * FROM Table_name;

Eg. SELECT * FROM tbl_user;

10. Selecting data using condition

Syntax: SELECT column1, column2,...
FROM table_name
WHERE condition;

Eg. SELECT uname, upassword FROM tbl_user WHERE uid=2;

11. Updating data (modifying the existing record in a table)

Syntax: UPDATE table_name
SET column1=value1 , column2=value2
WHERE condition;
[:. WHERE condition is optional]

Eg. UPDATE tbl_user SET uname='Ram'
WHERE uid=01;

12. Deleting data

Syntax: DELETE FROM table_name WHERE condition;

Eg. DELETE FROM tbl_user WHERE uname='Ram';

Write a C# program to make simple calculator using asp.net with C#.

Calculator.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Calculator.aspx.cs"
Inherits="NabKshitijCollege.WebForm1" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <table width="600">
                <tr><th width="300">Enter your first Number</th><td
width="300"><asp:TextBox ID="TextBox1" runat="server"></asp:TextBox></td></tr>
                <tr><th width="300">Enter your Second Number</th><td width="300">
                    <asp:TextBox ID="TextBox2" runat="server"></asp:TextBox></td></tr>
                <tr><th width="300"></th><td width="300"></td></tr>
                <tr><th width="300"></th><td width="300"><asp:Label ID="result"
runat="server" Text="Result"></asp:Label>
                    </td></tr>
                <tr><th width="300"></th><td width="300"></td></tr>
                <tr><th width="300">
                    </th><td width="300">
                        <asp:Button ID="Button1" runat="server" Text="Sum"
OnClick="Button1_Click" />&nbsp;&nbsp;&nbsp;<asp:Button ID="Button2" runat="server"
Text="Sub" OnClick="Button2_Click" />&nbsp;&nbsp;&nbsp;<asp:Button ID="Button3"
runat="server" Text="Mul" OnClick="Button3_Click" />&nbsp;&nbsp;&nbsp;<asp:Button
ID="Button4" runat="server" Text="Div" OnClick="Button4_Click" />
                    </td></tr>
                </table>
            </div>
        </form>
    </body>
</html>
```

Calculator.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
```

```

namespace NabKshitijCollege
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {

        }
        protected void Button1_Click(object sender, EventArgs e)
        {
            int res = int.Parse(TextBox1.Text) + int.Parse(TextBox2.Text);
            result.Text = res.ToString();
        }
        protected void Button2_Click(object sender, EventArgs e)
        {
            int res = int.Parse(TextBox1.Text) - int.Parse(TextBox2.Text);
            result.Text = res.ToString();
        }
        protected void Button3_Click(object sender, EventArgs e)
        {
            int res = int.Parse(TextBox1.Text) * int.Parse(TextBox2.Text);
            result.Text = res.ToString();
        }
        protected void Button4_Click(object sender, EventArgs e)
        {
            int res = int.Parse(TextBox1.Text) / int.Parse(TextBox2.Text);
            result.Text = res.ToString();
        }
    }
}

```

Output:

The screenshot shows a web browser window with the address bar displaying `https://localhost:44358/Calculator`. The page content includes two labels: "Enter your first Number" and "Enter your Second Number". The first input field contains the value "5" and the second contains "3". Below these fields, the result "8" is displayed. At the bottom, there are four buttons labeled "Sum", "Sub", "Mul", and "Div".

Example program to display form data in another page.

RegistrationForm.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="RegistrationForm.aspx.cs"
Inherits="NabKshitijCollege.RegistrationForm" %>
```

```
<!DOCTYPE html>
```

```
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <center><h1>Registration Form</h1></center>
    <hr />
    <center>
        <table width="500">
            <form id="form1" runat="server">
                <div>
                    <tr><th></th><th></th></tr>
                    <tr><th></th><th></th></tr>
                    <tr><th></th><th></th></tr>
                    <tr><th></th><th></th></tr>
                    <tr>
                        <th width="200"><asp:Label runat="server" Text="First Name"
ID="Label1"></asp:Label></th> <td width="300"><asp:TextBox ID="fname"
runat="server"></asp:TextBox></td>
                    </tr>
                    <tr>
                        <th width="200"><asp:Label ID="Label2" runat="server" Text="Last
Name"></asp:Label></th><td width="300"><asp:TextBox ID="lname"
runat="server"></asp:TextBox></td>
                    </tr>
                    <tr>
                        <th width="200"><asp:Label ID="Label3" runat="server"
Text="Contact"></asp:Label></th><td width="300"><asp:TextBox ID="contact"
runat="server"></asp:TextBox></td>
                    </tr>
                    <tr><th></th><th></th></tr>
                    <tr><th></th><th></th></tr>
                    <tr>
                        <th width="200"></th><td width="300"><asp:Button ID="Button1"
runat="server" Text="Save" OnClick="Button1_Click" /></td>
                    </tr>
                </div>
            </form>
        </table></center>
    </body>
</html>
```

RegistrationForm.aspx.cs

```
using System;
```

```

using System.Collections.Generic;
using System.Linq; using
System.Web; using System.Web.UI;
using System.Web.UI.WebControls;

namespace NabKshitijCollege
{
    public partial class RegistrationForm : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {

        }

        protected void Button1_Click(object sender, EventArgs e)
        {
            Response.Redirect("Report.aspx?fname="+
this.fname.Text + "lname="+
this.lname.Text + "contact="+
this.contact.Text

                );
        }
    }
}

```

Report.aspx

```

<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Report.aspx.cs"
Inherits="NabKshitijCollege.Report" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div>

                <asp:Label ID="Label1" runat="server" Text="Label"></asp:Label>
                <asp:Label ID="Label2" runat="server" Text="Label"></asp:Label>

                <asp:Label ID="Label3" runat="server" Text="Label"></asp:Label>

        </div>
    </form>
</body>
</html>

```

Report.aspx.cs

```

using System;
using System.Collections.Generic;
using System.Linq; using

```

```

System.Web; using System.Web.UI;
using System.Web.UI.WebControls;

namespace NabKshitijCollege
{
    public partial class Report : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            Label1.Text = Request.QueryString["fname"];
            Label2.Text = Request.QueryString["lname"];
            Label3.Text = Request.QueryString["contact"];
        }
    }
}

```

Output:

https://localhost:44358/RegistrationForm

Registration Form

First Name	Ram
Last Name	Chaudhary
Contact	9818084521
	Save

https://localhost:44358/Report?fname=Ram&lname=Chaudhary&contact=9818084521

RamlnameChaudharycontact9818084521

Database Connection String:

Default.aspx

```

<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Default.aspx.cs"
Inherits="MyProject.Default" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.1/dist/css/bootstrap.min.css"
rel="stylesheet" integrity="sha384-
4bw+/aepP/YC94hEpVNVgiZdgIC5+VKNBQNGCHeKRN+PtmoHDEXuppvnDJzQIu9"
crossorigin="anonymous">
    <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.1/dist/js/bootstrap.bundle.min.js"
integrity="sha384-HwwvtgBNo3bZJJLYd8oVXjrBZt8cqVSpeBNS5n7C8IVInixGAoxmnlMuBnhbgrkm"
crossorigin="anonymous"></script>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <center><h1>Registration Form</h1></center>
            <hr />
            <center>
                <table border="0">
                    <tr><th>
                        <asp:Label ID="Label2" runat="server" Text="Enter your ID"
CssClass="form-label"></asp:Label></th><td>
                            <asp:TextBox ID="id" runat="server" CssClass="form-
control"></asp:TextBox></td></tr>
                    <tr><th>
                        <asp:Label ID="Label1" runat="server" Text="Name"
CssClass="form-label"></asp:Label></th><td>
                            <asp:TextBox ID="name" runat="server" CssClass="form-
control"></asp:TextBox></td></tr>
                    <tr><th>
                        <asp:Label ID="Label3" runat="server" Text="Address"
CssClass="form-label"></asp:Label></th><td>
                            <asp:TextBox ID="address" runat="server" CssClass="form-
control"></asp:TextBox></td></tr>
                    <tr><th>
                        <asp:Label ID="Label4" runat="server" Text="Contact"
CssClass="form-label"></asp:Label></th><td>
                            <asp:TextBox ID="contact" runat="server" CssClass="form-
control"></asp:TextBox></td></tr>
                    <tr><th>
                        <asp:Label ID="Label5" runat="server" Text="Email"
CssClass="form-label"></asp:Label></th><td>
                            <asp:TextBox ID="email" runat="server" CssClass="form-
control"></asp:TextBox></td></tr>
                    <tr><td></td><td></td></tr>
                    <tr><td colspan="2">
                        <asp:Button ID="Button1" runat="server" Text="Save"
CssClass="btn btn-primary" OnClick="Button1_Click"/></td></tr>
                </table>
            </center>
        </div>
    </form>
</body>
</html>

```

```

</form>
</body>
</html>

```

The screenshot shows a web browser window with the address bar displaying 'localhost:44334/Default.aspx'. The page title is 'Registration Form'. The form is a vertical stack of input fields with labels to their left: 'Enter your ID' (value: 02), 'Name' (value: Rajan Shrestha), 'Address' (value: Pokhara), 'Contact' (value: 9841980054), and 'Email' (value: rajan@gmail.com). A blue 'Save' button is located below the email field. The browser's developer tools are open on the left side of the window.

Default.aspx.cs

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using System.Data.Sql;
using System.Data.SqlClient;
using System.Xml.Linq;

namespace MyProject
{
    public partial class Default : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {

        }

        protected void Button1_Click(object sender, EventArgs e)
        {
            string ConnectionString = "Data Source=DESKTOP-HCLFEPE;Initial
Catalog=smc;Persist Security Info=True;User ID=sa;Password=sqlserver2014";

            using (SqlConnection con = new SqlConnection(ConnectionString))
            {
                con.Open();
                int ID = Convert.ToInt32(id.Text);
                string Name = name.Text;
                string Address = address.Text;
                string Contact = contact.Text;
                string Email = email.Text;
            }
        }
    }
}

```



```
string Query = "INSERT INTO employee (id,name, address, contact,
email) VALUES (@Id,@Name, @Address, @Contact, @Email)";
```

```
using (SqlCommand cmd = new SqlCommand(Query, con))
{
    cmd.Parameters.AddWithValue("@ID", ID);
    cmd.Parameters.AddWithValue("@Name", Name);
    cmd.Parameters.AddWithValue("@Address", Address);
    cmd.Parameters.AddWithValue("@Contact", Contact);
    cmd.Parameters.AddWithValue("@Email", Email);

    cmd.ExecuteNonQuery();
}
}
```

```
}
}
```

SQLQuery1.sql - DESKTOP-HCLFEPE.smc (sa (55))* - Microsoft SQL Server Management Studio

The screenshot displays the Microsoft SQL Server Management Studio interface. The title bar indicates the file is 'SQLQuery1.sql - DESKTOP-HCLFEPE.smc (sa (55))*'. The menu bar includes File, Edit, View, Project, Debug, Tools, Window, and Help. The toolbar contains various icons for file operations, query execution, and debugging. The 'Object Explorer' on the left shows the server 'DESKTOP-HCLFEPE (SQL Server 12.0)' with a tree view of databases, including 'bcu', 'ReportServer', 'ReportServerTempDB', 'SHIKSHYANew', and 'smc'. The main query editor window shows the query 'SELECT * FROM employee;'. Below the editor, the 'Results' pane displays the execution output as a table with 5 columns: id, name, address, contact, and email. The table contains two rows of data.

	id	name	address	contact	email
1	1	saroj singh	kathamndu	9841980039	tosarojsingh2012@gmail.com
2	2	Rajan Shrestha	Pokhara	9841980054	rajan@gmail.com

Q. Write a server site script to performs a simple query on the **Librarydb** database that save the book as user want into **Book table** and display all records and also perform UPDATE and DELETE operation on the database?

For Insertion Data into the database

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Data;
using System.Data.SqlClient;

namespace BookApp
{
    class Program
    {
        static void Main(string[] args)
        {
            SqlConnection sc = new SqlConnection("Data Source=.;Initial
            Catalog=LibraryDB;Integrated Security=True");
            sc.Open();
            int id;
            string Isbn;
            string title;
            DateTime pdate;
            Console.WriteLine("Do u want add book record:");
            char ch = char.Parse(Console.ReadLine());
            while (ch == 'y' || ch == 'Y')
            {
                Console.WriteLine("Enter Book ID:");
                id = int.Parse(Console.ReadLine());
                Console.WriteLine("Enter Book ISBN:");
                Isbn = Console.ReadLine();
                Console.WriteLine("Enter Book Title:");
                title = Console.ReadLine();
                Console.WriteLine("Enter Book PDate:");
                pdate = DateTime.Now;
                SqlCommand cmd = new SqlCommand(@"insert into book_tbl
                values(@BID,@ISBN,@Title,@Pdate)", sc);
```

```

        cmd.Parameters.Add(new SqlParameter("BID",id));
        cmd.Parameters.Add(new SqlParameter("ISBN", Isbn));
        cmd.Parameters.Add(new SqlParameter("Title",title));
        cmd.Parameters.Add(new SqlParameter("Pdate", pdate));
        cmd.ExecuteNonQuery();
        Console.WriteLine("Do u want add book record:");
        ch = char.Parse(Console.ReadLine());
    }

    sc.Close();
    Console.WriteLine("insertion completed !!!");
    Console.ReadKey();
}
}
}

```

For Display Record from Database:

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Data;
using System.Data.SqlClient;

namespace BookApp
{
    class Program
    {
        static void Main(string[] args)
        {
            SqlConnection sc = new SqlConnection("Data Source=.;Initial
Catalog=LibraryDB;Integrated Security=True");
            sc.Open();
            int id;
            string Isbn;
            string title;
            DateTime pdate;

            SqlCommand cmd = new SqlCommand(@"select * from book_tbl",sc);
            SqlDataReader rdr = cmd.ExecuteReader();
            while(rdr.Read())
            {
                id = (int)rdr["BID"];
                Console.Write(id);
                Console.Write(" ");
                Isbn = (string)rdr["ISBN"];

```

```

        Console.Write(Isbn);
        Console.Write(" ");
        title= (string)rdr["Title"];
        Console.Write(title);
        Console.Write(" ");
        pdate= (DateTime)rdr["Pdate"];
        Console.Write(pdate);
        Console.WriteLine();
    }
    sc.Close();
    Console.ReadKey();
}
}
}

```

For delete data from database:

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Data;
using System.Data.SqlClient;

namespace BookApp
{
    class Program
    {
        static void Main(string[] args)
        {
            SqlConnection sc = new SqlConnection("Data Source=.;Initial
Catalog=LibraryDB;Integrated Security=True");
            sc.Open();

            SqlCommand cmd = new SqlCommand(@"delete from book_tbl where BID='1'", sc);
            cmd.ExecuteNonQuery();
            Console.WriteLine("Deleted !!!");

            sc.Close();
            Console.ReadKey();
        }
    }
}

```

For update data in database:

```

using System;
using System.Collections.Generic;

```

```
using System.Linq;
using System.Text;
using System.Data;
using System.Data.SqlClient;
```

```
namespace BookApp
```

```
{
```

```
    class Program
```

```
    {
```

```
        static void Main(string[] args)
```

```
        {
```

```
            SqlConnection sc = new SqlConnection("Data Source=.;Initial
Catalog=LibraryDB;Integrated Security=True");
            sc.Open();
```

```
            SqlCommand cmd = new SqlCommand(@"update book_tbl set ISBN='QW-56-TR' WHERE
            BID= '3'", sc);
```

```
            cmd.ExecuteNonQuery();
```

```
            Console.WriteLine("UPDATED !!!");
```

```
            sc.Close();
```

```
            Console.ReadKey();
```

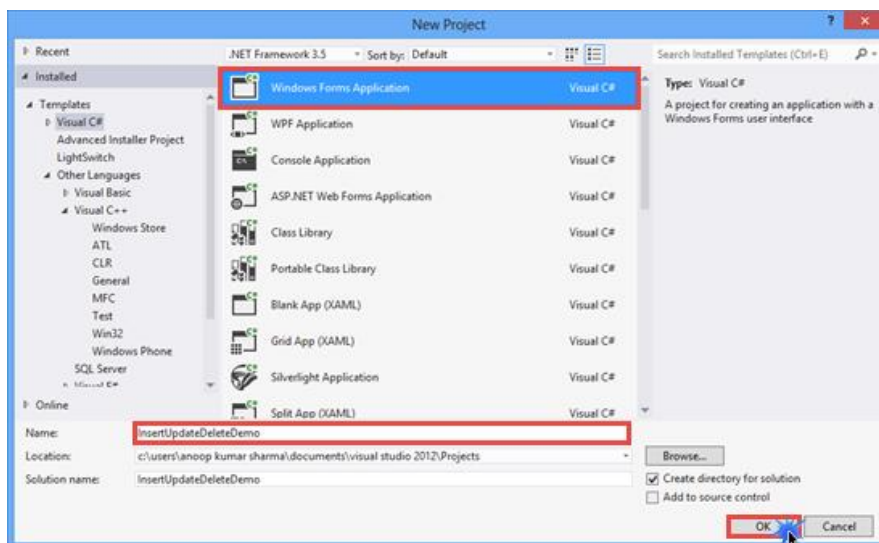
```
        }
```

```
    }
```

```
}
```

How to insert, update and delete records in a DataGridView in a C# Windows Forms application.

1. Create a new Windows Forms application.



2. Create a database (named Sample). Add a table tbl_Record. The following is the table schema for creating tbl_Record.

ANOOP.Sample - dbo.tbl_Record			
	Column Name	Data Type	Allow Nulls
▶	ID	int	☐
	Name	varchar(50)	☒
	State	varchar(50)	☒
			☐

3. Create a form (named frmMain) and drop a Label, TextBox, Button and DataGridView control from the ToolBox.

Now, go to the frmMain.cs code and add the System.Data and System.Data.SqlClient namespaces.

frmMain.cs Code

```

1. using System;
2. using System.Data;
3. using System.Windows.Forms;
4. using System.Data.SqlClient;
5.
6. namespace InsertUpdateDeleteDemo
7. {
8.     public partial class frmMain : Form
9.     {
10.         SqlConnection con= new SqlConnection("Data Source=.;Initial Catalog=Sample;Integrated Security=true;");

```

```

11.      SqlCommand cmd;
12.      SqlDataAdapter adapt;
13.      //ID variable used in Updating and Deleting Record
14.      int ID = 0;
15.      public frmMain()
16.      {
17.          InitializeComponent();
18.          DisplayData();
19.      }
20.      //Insert Data
21.      private void btn_Insert_Click(object sender, EventArgs e)
22.      {
23.          if (txt_Name.Text != "" && txt_State.Text != "")
24.          {
25.              cmd = new SqlCommand("insert into tbl_Record(Name,State) values(@name,@
state)", con);
26.              con.Open();
27.              cmd.Parameters.AddWithValue("@name", txt_Name.Text);
28.              cmd.Parameters.AddWithValue("@state", txt_State.Text);
29.              cmd.ExecuteNonQuery();
30.              con.Close();
31.              MessageBox.Show("Record Inserted Successfully");
32.              DisplayData();
33.              ClearData();
34.          }
35.          else
36.          {
37.              MessageBox.Show("Please Provide Details!");
38.          }
39.      }
40.      //Display Data in DataGridView
41.      private void DisplayData()
42.      {
43.          con.Open();
44.          DataTable dt=new DataTable();
45.          adapt=new SqlDataAdapter("select * from tbl_Record",con);
46.          adapt.Fill(dt);
47.          dataGridView1.DataSource = dt;
48.          con.Close();
49.      }
50.      //Clear Data
51.      private void ClearData()
52.      {
53.          txt_Name.Text = "";
54.          txt_State.Text = "";
55.          ID = 0;
56.      }
57.      //dataGridView1 RowHeaderMouseClicked Event
58.      private void dataGridView1_RowHeaderMouseClicked(object sender, DataGridViewCellM
ouseEventArgs e)
59.      {
60.          ID = Convert.ToInt32(dataGridView1.Rows[e.RowIndex].Cells[0].Value.ToString
());
61.          txt_Name.Text = dataGridView1.Rows[e.RowIndex].Cells[1].Value.ToString();
62.          txt_State.Text = dataGridView1.Rows[e.RowIndex].Cells[2].Value.ToString();
63.      }
64.      //Update Record
65.      private void btn_Update_Click(object sender, EventArgs e)
66.      {
67.          if (txt_Name.Text != "" && txt_State.Text != "")

```

```

68.         {
69.             cmd = new SqlCommand("update tbl_Record set Name=@name,State=@state where ID=@id", con);
70.             con.Open();
71.             cmd.Parameters.AddWithValue("@id", ID);
72.             cmd.Parameters.AddWithValue("@name", txt_Name.Text);
73.             cmd.Parameters.AddWithValue("@state", txt_State.Text);
74.             cmd.ExecuteNonQuery();
75.             MessageBox.Show("Record Updated Successfully");
76.             con.Close();
77.             DisplayData();
78.             ClearData();
79.         }
80.         else
81.         {
82.             MessageBox.Show("Please Select Record to Update");
83.         }
84.     }
85.     //Delete Record
86.     private void btn_Delete_Click(object sender, EventArgs e)
87.     {
88.         if(ID!=0)
89.         {
90.             cmd = new SqlCommand("delete tbl_Record where ID=@id",con);
91.             con.Open();
92.             cmd.Parameters.AddWithValue("@id",ID);
93.             cmd.ExecuteNonQuery();
94.             con.Close();
95.             MessageBox.Show("Record Deleted Successfully!");
96.             DisplayData();
97.             ClearData();
98.         }
99.         else
100.        {
101.            MessageBox.Show("Please Select Record to Delete");
102.        }
103.    }
104. }
105.

```

When the user clicks on the Row Header of a row then the data present in the cell of the row is stored into the TextBoxes. The DisplayData() method fills in the data in the DataGridView.

The Clear() method clears the data present in the TextBox as well as in the ID(int) variable.

Final Preview

Form1

Name

State

	ID	Name	State
▶	1	Anoop Sharma	Delhi
	2	Ankit Pandey	Bihar
	3	Gaurav Kapoor	Mumbai
	4	Arjit Singh	Punjab
*			